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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 5

Title: Building Sales Performance Dashboards

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Experiment No.: 5

Building Sales Performance Dashboards

Problem Statement

You are a Sales Analyst for a retail chain. The sales director wants to monitor revenue, profit, product performance, and regional sales through a single dashboard. Design an interactive sales dashboard that supports data-driven decision-making.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the Walmart sales dataset in CSV format.
- Verified that the dataset contained sales, store, holiday, fuel price, temperature, CPI, and unemployment information.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported the Walmart CSV dataset using the **Get Data** option.
- Loaded the dataset into Power BI for analysis.

Step 3: Data Cleaning and Transformation

- Checked for missing values and duplicate records.
- Verified and corrected data types for all columns.
- Created additional columns including **Year**, **Month**, **Quarter**, and **Holiday Type** for analysis.
- Formatted date columns and sorted month names using the Month Number column.

Step 4: Data Modeling

- Verified relationships and data integrity within the dataset.
- Created DAX measures for:
 - Total Weekly Sales
 - Average Weekly Sales
 - Best Performing Store
 - Worst Performing Store
 - Holiday Sales

Step 5: Dashboard Development

Created KPI cards for:

- Total Weekly Sales
 - Average Weekly Sales
 - Best Performing Store
 - Worst Performing Store
 - Holiday Sales
- Added slicers for: ☐ Store ☐ Year ☐ Month ☐ Holiday Type
- Applied formatting, color themes, titles, and interactive filters to create a professional dashboard.

Output :

New Measures using “DAX”:

Table: Walmart (6,435 rows) Column: Measure (0 distinct values)

Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
2	02-07-2010	2003940.64	0	82.74	2.669	210.8803726	8.099
2	09-07-2010	1880902.62	0	82.59	2.642	210.7657317	8.099
2	16-07-2010	1845879.79	0	85.32	2.623	210.7577954	8.099
2	23-07-2010	1781717.71	0	87.66	2.608	210.8921319	8.099
2	30-07-2010	1804246.16	0	83.49	2.64	211.0264684	8.099
2	06-08-2010	1991909.98	0	89.53	2.627	211.1608049	8.099
2	13-08-2010	1895601.05	0	89.05	2.692	211.2951413	8.099
2	20-08-2010	1964335.23	0	88.7	2.664	211.2596586	8.099
2	27-08-2010	1863840.49	0	87.12	2.619	211.2241759	8.099
2	03-09-2010	1904608.09	0	81.83	2.577	211.1886931	8.099
2	17-09-2010	1793903.6	0	82.05	2.582	211.1806415	8.099
2	24-09-2010	1724557.22	0	81.79	2.624	211.2552578	8.099
15	02-07-2010	709337.11	0	66.13	2.958	132.5218667	8.099
15	09-07-2010	691497.62	0	77.33	2.94	132.5461333	8.099
15	16-07-2010	63203.69	0	72.83	2.933	132.5667742	8.099
15	23-07-2010	598553.43	0	73.2	2.924	132.5825806	8.099
15	30-07-2010	619224.06	0	72.04	2.932	132.5983871	8.099
15	06-08-2010	639651.24	0	72.7	2.942	132.6141935	8.099
15	13-08-2010	622437.08	0	72.17	2.923	132.63	8.099
15	20-08-2010	637090.44	0	71.1	2.913	132.6616129	8.099
15	27-08-2010	649791.15	0	65.07	2.885	132.6932258	8.099
15	03-09-2010	638647.21	0	73.17	2.86	132.7248387	8.099
15	17-09-2010	583210.87	0	57.94	2.846	132.7670667	8.099
15	24-09-2010	548542.47	0	62.53	2.837	132.7619333	8.099
19	02-07-2010	1549018.68	0	66.25	2.958	132.5218667	8.099
19	09-07-2010	1577541.24	0	78.22	2.94	132.5461333	8.099
19	16-07-2010	1412157.02	0	74.6	2.933	132.5667742	8.099
19	23-07-2010	1377945.91	0	74.6	2.933	132.5667742	8.099

Dashboard:



An interactive Sales Performance Dashboard was successfully developed in Power BI using the Walmart sales dataset. The dashboard provides an overview of weekly sales performance through KPI cards and multiple interactive visualizations. Users can analyze sales trends over time, compare store performance, evaluate holiday and non-holiday sales, observe the relationship between temperature and sales, identify store rankings, and understand monthly sales changes. Interactive slicers for Store, Year, Month, and Holiday Type enable dynamic filtering, making the dashboard user-friendly and effective for business analysis.

Conclusion

The Sales Performance Dashboard was successfully designed and implemented in Power BI. The dashboard combines KPI cards, interactive charts, and slicers to present meaningful business insights from the Walmart sales dataset. It enables users to monitor sales trends, compare store performance, evaluate seasonal and holiday impacts, and make informed business decisions. Overall, the dashboard transforms raw sales data into a clear, interactive, and decision-oriented business report.

Learning Outcomes

After completing this experiment, I learned:

- How to import and prepare business data in Power BI.
- How to clean and transform sales data for analysis.
- How to create DAX measures for KPI calculations.
- How to develop interactive KPI cards and business dashboards.
- How to create Line, Column, Scatter, Ribbon, Treemap, Donut, and Waterfall charts.
- How to use slicers for interactive dashboard filtering.
- How to apply professional formatting and dashboard design principles.
- How to analyse sales trends, store performance, and holiday sales using Power BI visualisations.
- How to present business insights that support data-driven decision-making.

GitHub Repository:

<https://github.com/deepakMudiraj06/Power-BI/tree/main/Assignment%205>